9A Lesson 6

Genes, adaptation and extinction

Year 9 Science · 9A Genetics and evolution · Student Book pages 16 to 17

Name _____ Class _____ Date _____

Today I can:

☐ I can explain how changes in abiotic and biotic factors can make a species endangered or extinct, using the idea of adaptations.

☐ I can describe ways of preserving biodiversity, including nature reserves, breeding programmes and gene banks.

Key words

adaptation a characteristic that helps an organism survive in its habitat

endangered a species in danger of becoming extinct

extinct a species with no living members left

abiotic factor a non-living part of the environment, e.g. rainfall, temperature

biotic factor the activities of other organisms, e.g. competition, disease, hunting

competition when organisms need the same limited resource, e.g. food or light

native species a species that is naturally found in a part of the world

biodiversity the number of different species in an area or on Earth

nature reserve an area where habitats and species are protected

breeding programme breeding endangered animals or plants in captivity to increase their numbers

gene bank a store of seeds, gametes or cells kept cold so a species can be grown again

● Core Everyone does this section.

1. Match each scientific term to its definition. Draw a line.

environment	a non-living (physical) part of the surroundings, e.g. temperature
environmental factor	the activities of other organisms, e.g. competition, predation
abiotic factor	an organism's surroundings
biotic factor	differences caused by the surroundings
environmental variation	a part of the surroundings that can change an organism

2. Complete the sentences using words from the box. You may use each word once, more than once or not at all.

Word box: abiotic adaptations biotic camouflaged characteristics common desert endangered environmental forest habitats inherited

Animals have _____ to help them survive in their _____ . These adaptations are _____ characteristics. For example, the fur of sand cats matches the colour of the sand in a _____ habitat. The cats are _____ as they hunt gerbils. If more rain fell and the desert got greener, the sand cats would not be _____ and may start to become _____ in that area.

3. Tick one box for each question.

a) What is a native species?

☐ A A species that is only active at night.

☐ B A species that is eaten by people in a certain part of the world.

☐ C A species that is in danger of becoming extinct.

☐ D A species that is naturally found in one part of the world.

b) Wolves and foxes compete for the same prey. What type of competition is this?

- ☐ A inter-specific (between species)
- ☐ B intra-specific (within one species)
- ☐ C counter-specific
- ☐ D inner-specific

4. Give three resources that plants compete for.

●● Stretch When Core is finished.

S1. Grey squirrels were brought to the UK in the 1870s. Red squirrels are now rare.

a) Explain why a squirrel population might decrease during a long winter.

b) Explain how grey squirrels are better adapted to long winters than reds.

c) Explain how genes are responsible for this difference.

S2. Conifer trees have smaller seeds and thinner, more flexible branches than trees with broad leaves. Suggest why red squirrels may be better adapted to conifer forests than greys.

S3. Sundews are plants adapted to wet, acidic bogs. Most have only very small roots because a lot of water is available. Explain why sundews may not continue to survive in an area where the average temperature starts to increase.

●●● Challenge Explain in full sentences.

C1. About 1500 years ago, giant flightless moa birds lived in New Zealand. They were the prey of huge Haast's eagles. When humans arrived, they hunted the moa to extinction.

a) Explain an adaptation that Haast's eagles might have had for hunting moa.

b) Suggest and explain why Haast's eagles became extinct soon after the moa.

C2. Explain how a gene bank can allow a plant species to survive if it becomes extinct in the wild, and give one reason why a gene bank alone is not enough to save a species.

My thinking (any language)

Red squirrels died out in some parts of the UK in the 1980s and there are plans to reintroduce them. What could be done to help the reintroduced red squirrels survive?

Word bank adaptation · endangered · extinct · abiotic factor · biotic factor · competition · native species · biodiversity · nature reserve · gene bank

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Exit ticket 9A Lesson 6: Genes, adaptation and extinction Name _____

1. What does it mean if a species is endangered?

2. Give one biotic factor that can cause a species to become extinct.

3. Give one way of preserving biodiversity and say how it works.

Today I can: Goal 1 ☐ Goal 2 ☐